

Into the Future of Coding with GitHub Copilot

SkillUp **AI**

Boosting Productivity for Devs &
Data Scientists with AI



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Data & Applied Scientist II
Web Experiences Team

A little about me...

- **Microsoft** | **Bing Autosuggest** (2021 – Present)
 - Data & Applied Scientist II (2023 - Present)
 - Data & Applied Scientist (2021 - 2023)
- **IIT Bombay** (2017 – 21)
 - Major in Mechanical Engineering (9.90 CPI)
 - Minor in Computer Science & Engineering
 - Department Rank 1 | Institute Rank 2
- Amazon #1 **Best Selling Author** of "**Beyond Code**"
- Co-Author of AI Newsletter – **The Vision, Debugged;**
- **AI & Data Science Mentor** @Topmate (2023 – Present)
- Holder of 1 **US Patent**
- Areas of Interest: AI Tools, NLP, Data Analysis



Agenda

- **Understanding AI in Coding**
- **What is GitHub Copilot?**
- **Getting Started with GitHub Copilot**
- **GitHub Copilot in Action!**
 - **Code Generation**
 - **Code Explanation**
 - **Optimization**
 - **Debugging**
 - **Testing**
 - **Documentation**
 - **... and much more!**
- **Q&A**

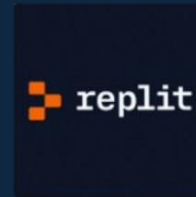
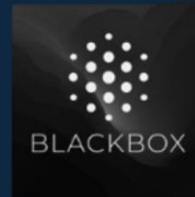


Are we Ready to Code Smarter with AI?



Understanding AI in Coding

- AI has been dominating every industry, including software dev & data science
 - New models, tools & capabilities launched almost every month!



- Professionals are using AI as their collaborator for massive productivity gains!

80.5%

Programmers are
using AI tools for
learning

58.5%

Programmers use AI
tools for writing code

> 30%

Part of the job where
AI helps coders

36.6%

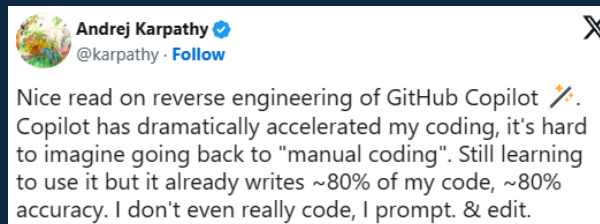
Programmers haven't
onboarded due to
learning curve

What is GitHub Copilot?



Introducing Your AI Pair Programmer - GitHub Copilot

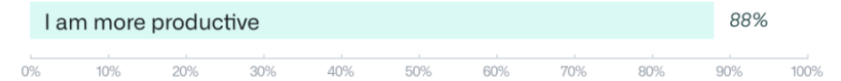
- One of the most widely adopted AI Developer Tools
 - **37,000+** business have adopted it
 - **1 in 3** Fortune 500 Companies use it
- Developers using GitHub Copilot complete tasks **55% faster** than others
- Features:
 - Works with multiple languages
 - AI-powered Real-time Suggestions
 - Code Completion
 - Analysis & Explanation
 - Testing & Debugging
 - Automatic Documentation
 - Conversations with Codebase



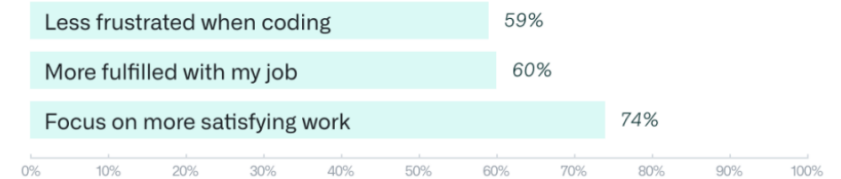
Andrej Karpathy's tweet about using GitHub Copilot for most of his job now

When using GitHub Copilot...

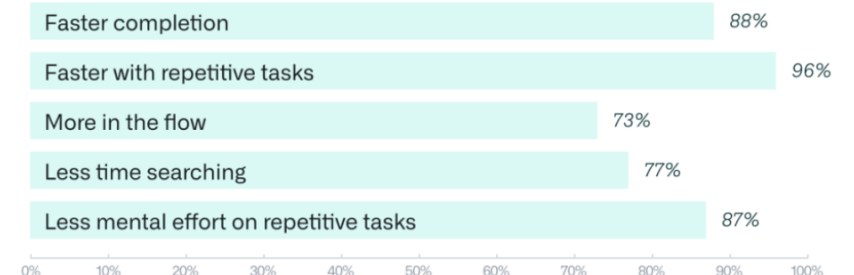
Perceived Productivity



Satisfaction and Well-being*



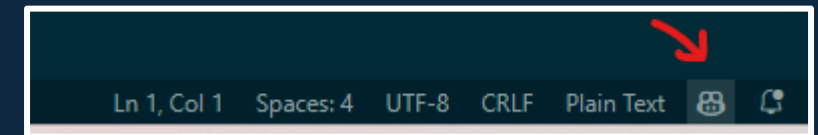
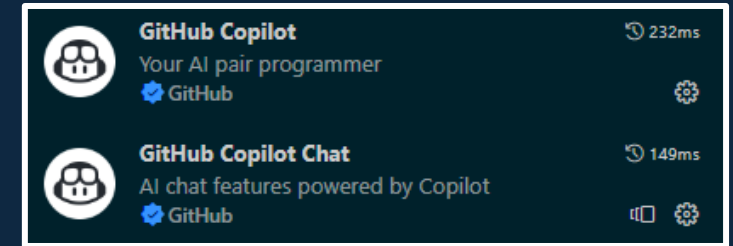
Efficiency and Flow*



Survey responses measuring dimensions of developer productivity when using GitHub Copilot

Getting Started with GitHub Copilot

- Go to <https://aka.ms/copilot> & follow the steps
- Available as an extension in your preferred environment
 - Visual Studio Code Extensions
 - GitHub Copilot: AI-powered code generation
 - GitHub Copilot Chat: Fully integrated AI chat experience
- Sign into GitHub from VS Code for authorization
- Get your first suggestion as you start typing:
 - Accept full suggestion: Tab
 - See alternative suggestions: Alt +]
 - Partially accept suggestion: Ctrl + →
- If disabled, click on the show copilot context menu to enable the extension



Time for some Live Action!



Code Generation



Code Explanation



Optimization



Debugging



Testing



Documentation



Here's What We've Learned...

- How **AI-powered tools** are helping developers skyrocket their productivity
- **What** is GitHub Copilot & what are the **features** it offers
- How to install & start using **GitHub Copilot in VS Code**
- How to use real-time **code generation capabilities** of GitHub Copilot
 - Writing functions, algorithms, perform data analysis & visualization
- How to leverage **code-understanding capabilities** of GitHub Copilot
 - Explaining code, optimization & debugging
- How to automate mundane tasks like **text processing, writing tests & documentation**
- How to **learn about new technical topics**

[GitHub Repo](#)

That's All Folks!

For More Interesting Stuff on AI , Data Science & the Latest Technologies...

